AI-Based Archaeological Pottery Fragment Matching



From a plan to a working, evaluated system: appearance + fracture-edge geometry on the Daxinzhuang-18K benchmark

PRT LLT · 21CSE451T

Review 2

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PROGRESS SINCE REVIEW 1

From plan to evidence

REVIEW 1

Dataset named, schema to verify, pipeline planned, no results

REVIEW 2

Data acquired and verified, pipeline built, 5 experiments, working demo

124
real sherds in the test set

111
expert-confirmed joins

16
model configurations compared

5
experiments run (E1–E5)

- **Data**: Daxinzhuang-18K found on Zenodo; only the needed parts fetched (range requests, CRC-checked)
- **Pipeline**: segmentation, CLAHE, contours, colour/LBP/ResNet-50/DINOv2 features
- **Geometry**: complementary-curvature edge matcher with Kabsch + ICP alignment
- **Model**: logistic-regression fusion, compared with RF and SVM
- **Evaluation**: leak-free vessel-grouped CV, bootstrap CIs, ablations
- **Prototype**: Gradio demo with ranked candidates and fit preview

DATASET

Daxinzhuang-18K, verified

Real photos	36,532 photos · 17,635 records	metadata + photo segmentation
Gold standard	124 sherds · 111 expert joins + poses	main train / test set
Simulated	50,736 synthetic break pairs	geometry sanity check (2,212 used)
Refit sheet	251 expert joining groups	ground truth, raw-photo test (next)

Zenodo 21553799 · Ji et al., 2026 · CC-BY 4.0

1.46%

of the 7,626 sherd pairs are joins: extreme class imbalance

33

vessel groups (2–11 sherds each), used to split train/test





METADATA UNDERSTANDING

What pixels cannot tell us

PHOTO ID ≠ SHERD ID

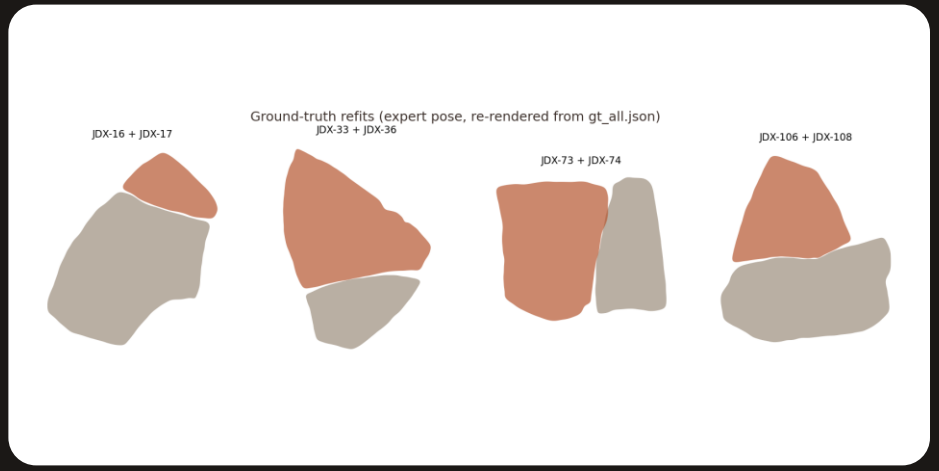
Sherd JD00001 → photos JD_00001 (exterior) + JD_00002 (interior). Mapping by number would pair the wrong photos. Fixed via the Exterior_Photo column.

POSE CONVENTIONS VERIFIED

Gold: recentre on alpha centroid, rotate, place. Simulated: image centre, opposite sign. Verified: true pairs touch 75–170 px with ≤7% overlap.

USED FOR

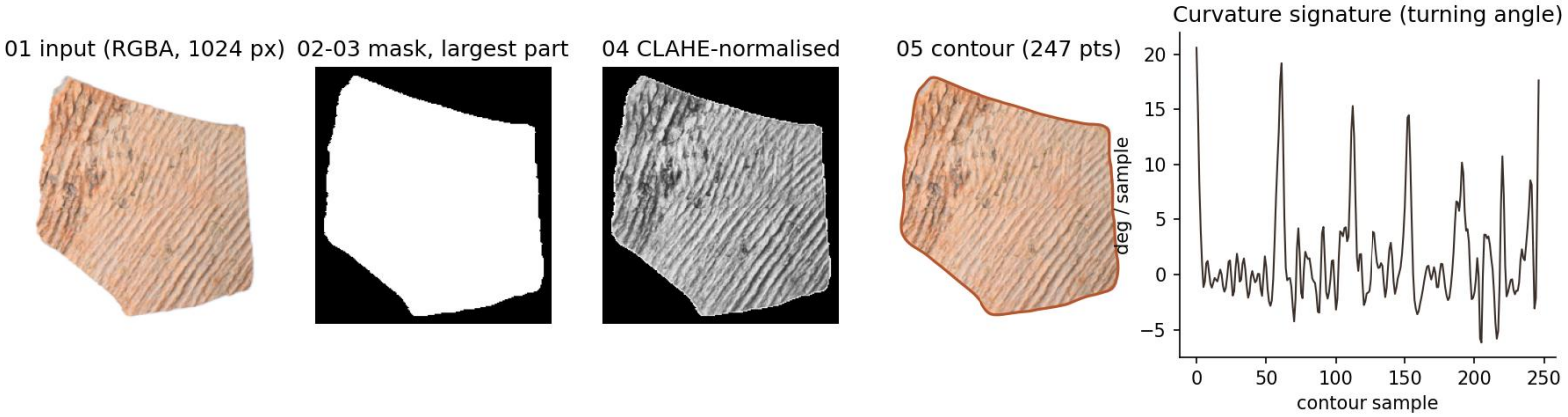
- **Vessel groups** → leak-free folds
- **Notes** (spalling) explain errors
- **Layer, part** → future priors
- Only 547 of 17,635 records typed: too sparse as a label

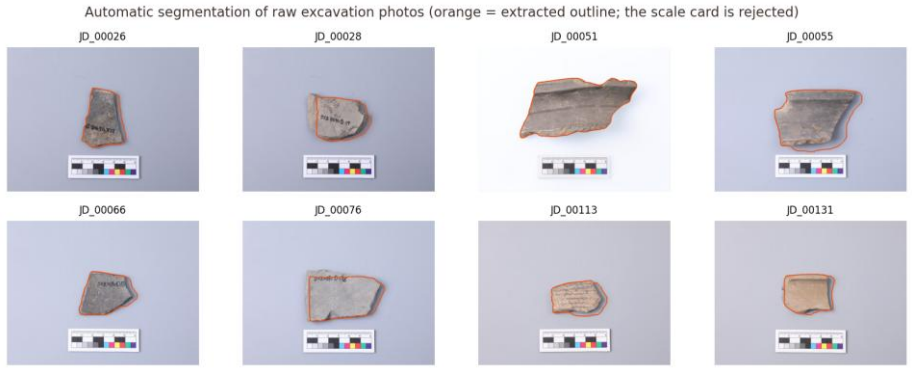


Expert joins re-rendered with our own pose code. Visible gaps along the joints are edge erosion, the central obstacle for geometry.

THE PROCESS, IMPLEMENTED

Preprocessing pipeline

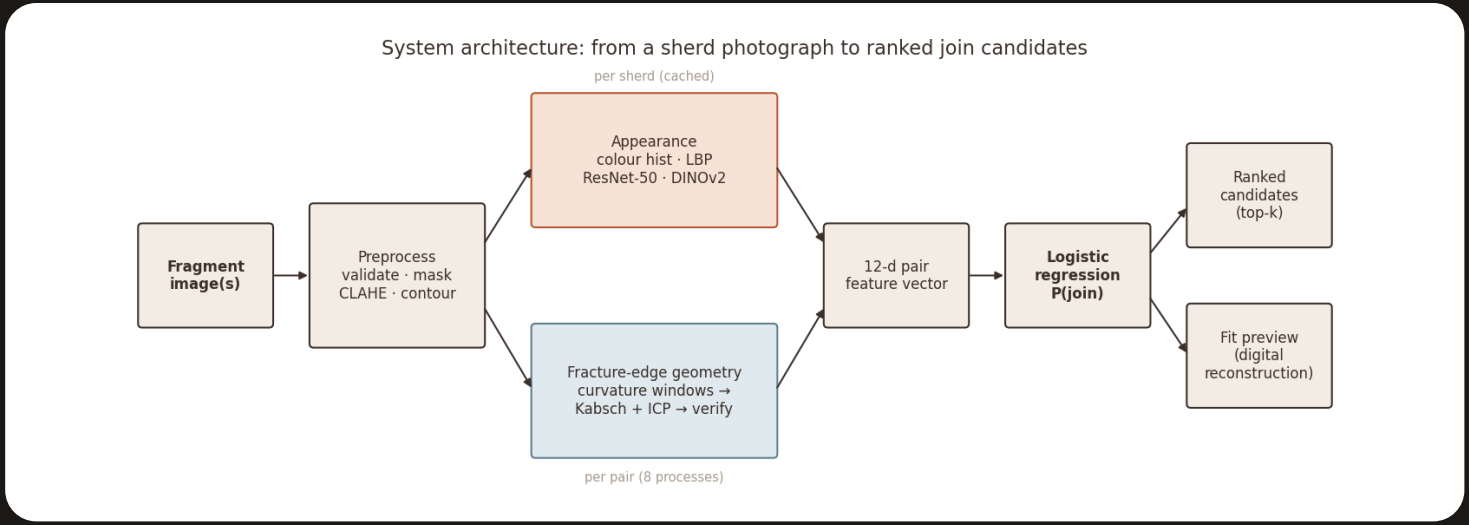




- Scale kept fixed: edge lengths must match
- Raw photos: Otsu → card filter → **GrabCut**
- Remaining error: light edges cut, shadows kept

MODEL SELECTION

Two cues, one interpretable classifier



4 appearance: colour histogram, LBP texture, ResNet-50, DINOv2

8 geometry: contact, contact fraction, overlap, rmse, fit score, stripe-angle gap (+confidence), area ratio

PATTERN TO RECOGNISE

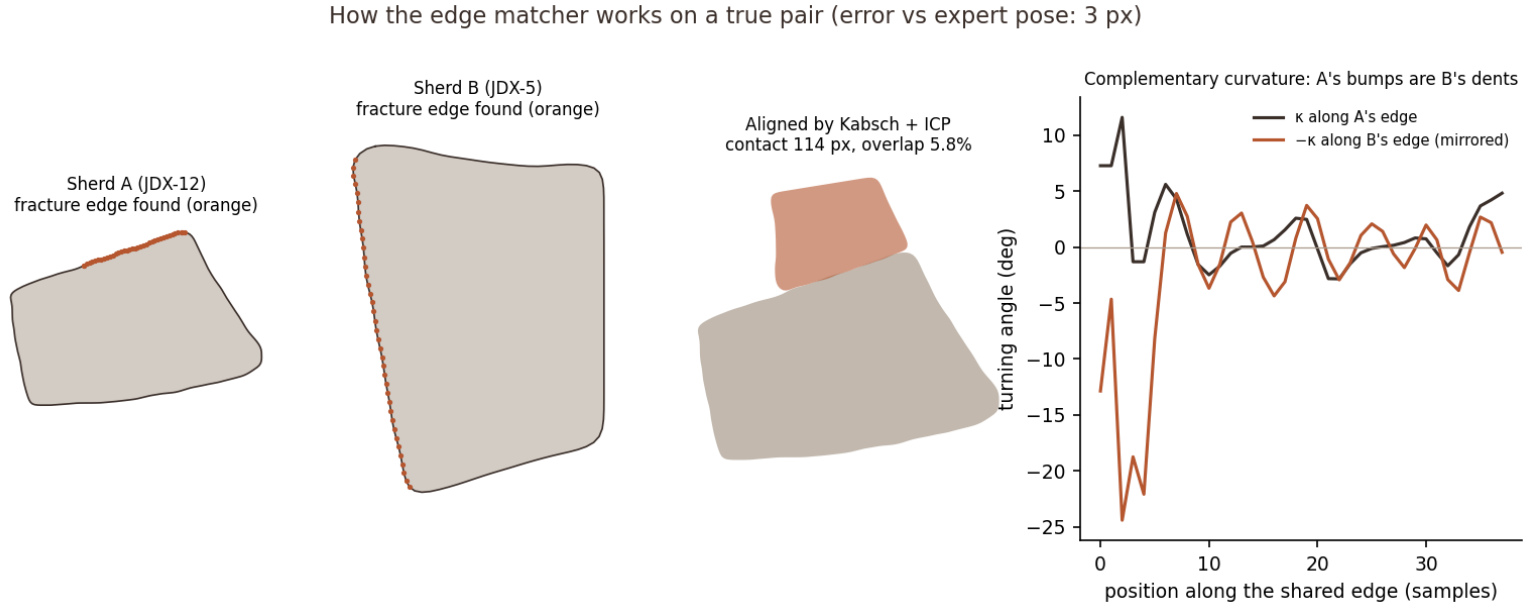
Does sherd pair (A, B) join? Used as ranking: sort the gallery by P(join).

LOGISTIC REGRESSION

- $P(\text{join}) = \sigma(w^T z + b)$
- 12 standardised pair features
- class-balanced, L2 (C = 1)

FRACTURE-EDGE GEOMETRY

Complementary curvature matching



01 Signature

turning angle every 3 px

02 Seeds

A vs reversed −B windows

03 Align

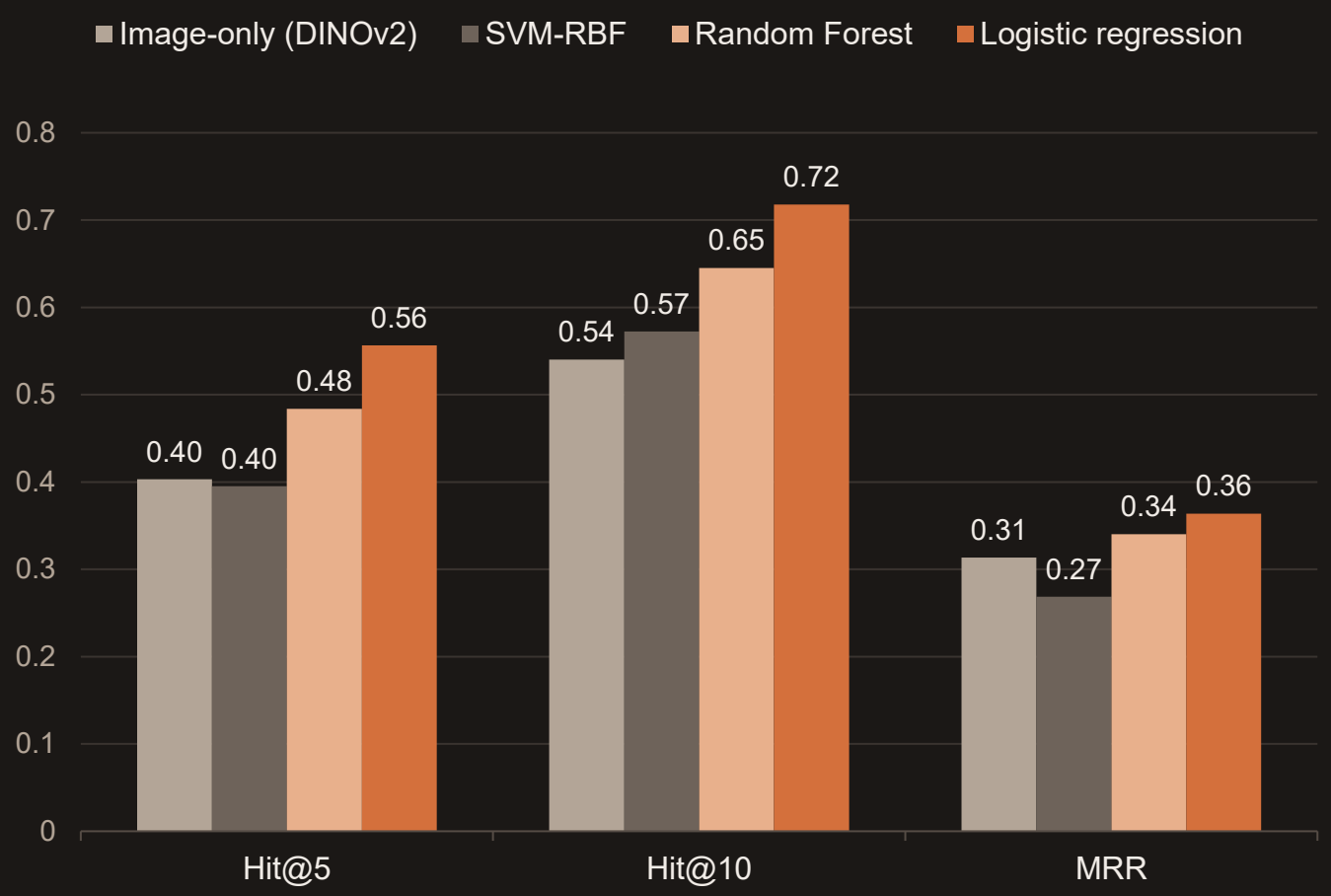
Kabsch, then ICP

04 Verify

contact ↑ overlap ↓ stripes agree

JUSTIFICATION

Why logistic regression



DATA SIZE

111 positives fit a 13-parameter model; non-linear models and deep nets overfit

INTERPRETABILITY

Signed weight per cue: experts can see why a sherd was suggested

COST AND RESULT

Milliseconds to train; best Hit@5 and Hit@10 of all classifiers

EXPERIMENTAL SETUP

Leak-free evaluation

VALIDATION

- 5-fold **GroupKFold** over the 33 vessel groups
- No vessel in both train and test
- Each query ranked against all 123 others
- F1 threshold from inner CV on training pairs

METRICS AND WHY

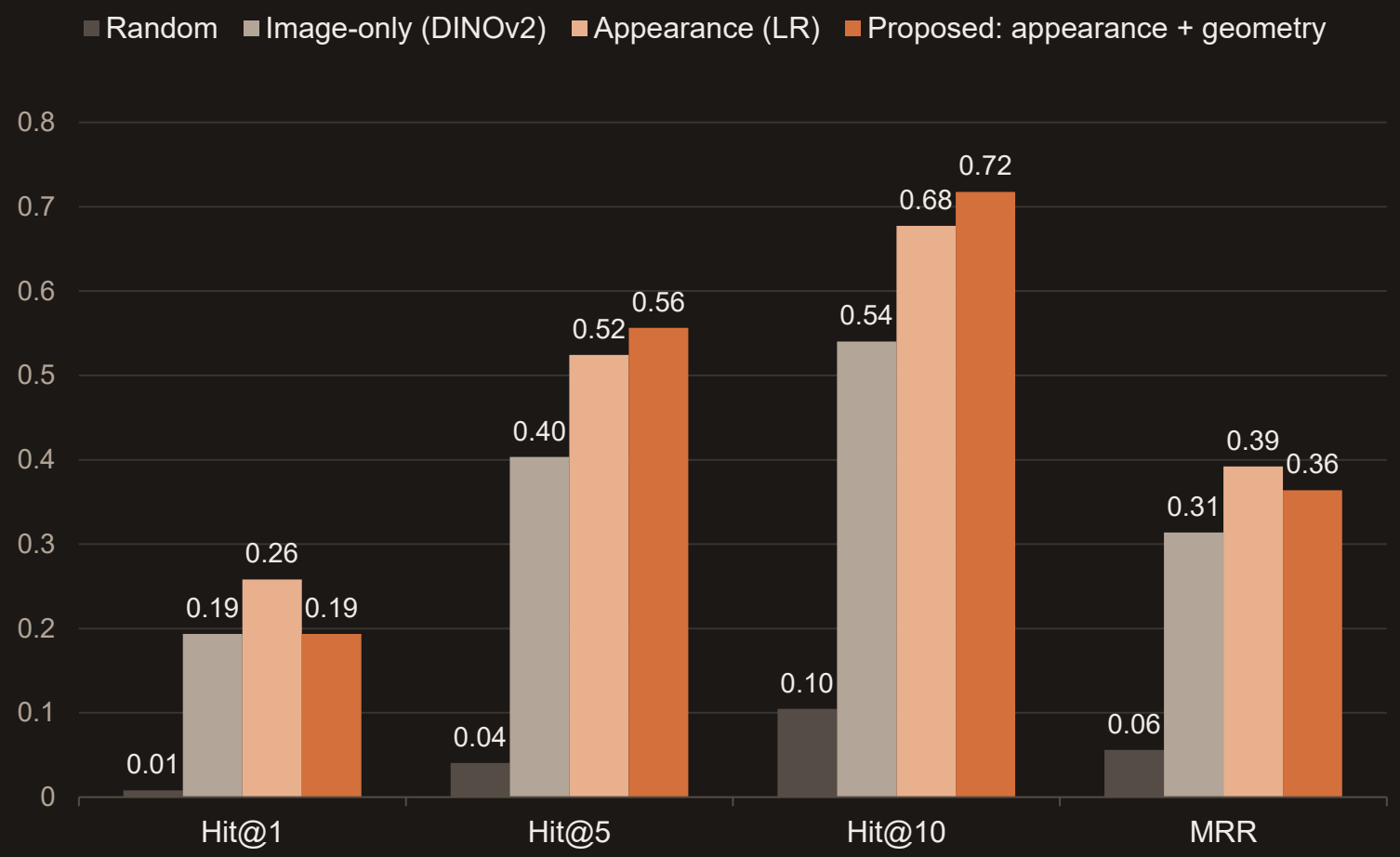
- **Hit@k**: what the expert checks
- **MRR, mAP**: rank quality
- **P / R / F1, PR-AUC**: pair decisions
- Accuracy is misleading at 1.46% positives
- **Bootstrap 95% CIs** (5,000×)

EXPERIMENTS

- E1 image-only vs fusion
- E2 LR vs RF vs SVM
- E3 ablation of every cue
- E4 geometry on synthetic breaks
- E5 pose accuracy on real joins
- E6 raw photos (next step)

RESULTS (E1)

Better than image-only retrieval



72%

of sherds have a true partner in the top 10

+17.7 pts

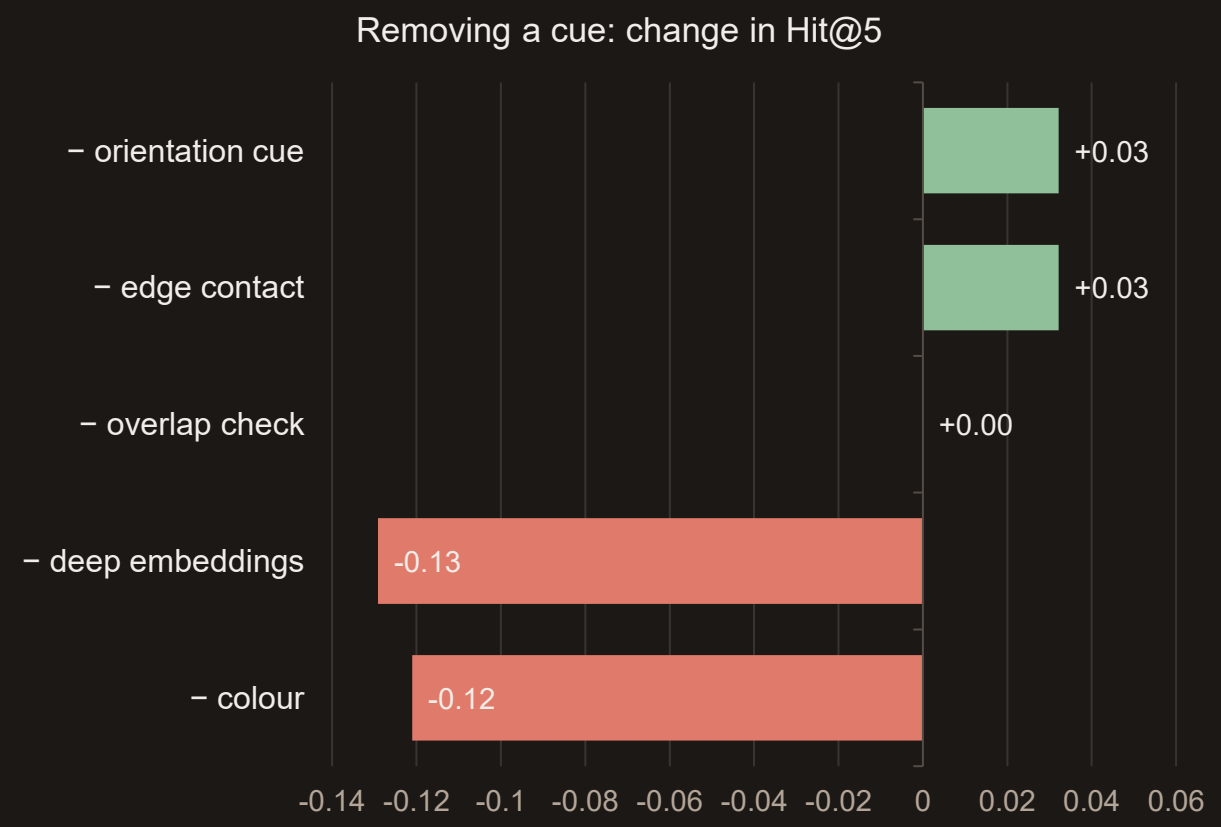
Hit@10 vs image-only (95% CI +9.7 to +26.6)

+15.3 pts

Hit@5 vs image-only (95% CI +5.6 to +25.0)

TESTING THE HYPOTHESIS

Does geometry pull its weight?



Red = the cue helped (removing it hurts). Green = removing it helps.

Hit@1	**-6.5**	95% CI -12.1 to -1.6 significant
Hit@5	**+3.2**	95% CI -2.4 to +9.7 not significant
Hit@10	**+4.0**	95% CI -2.4 to +10.5 not significant

Adding geometry to learned appearance: **H1 only partly supported**

DIAGNOSIS (E4, E5)

The bottleneck is the edge, not the matcher

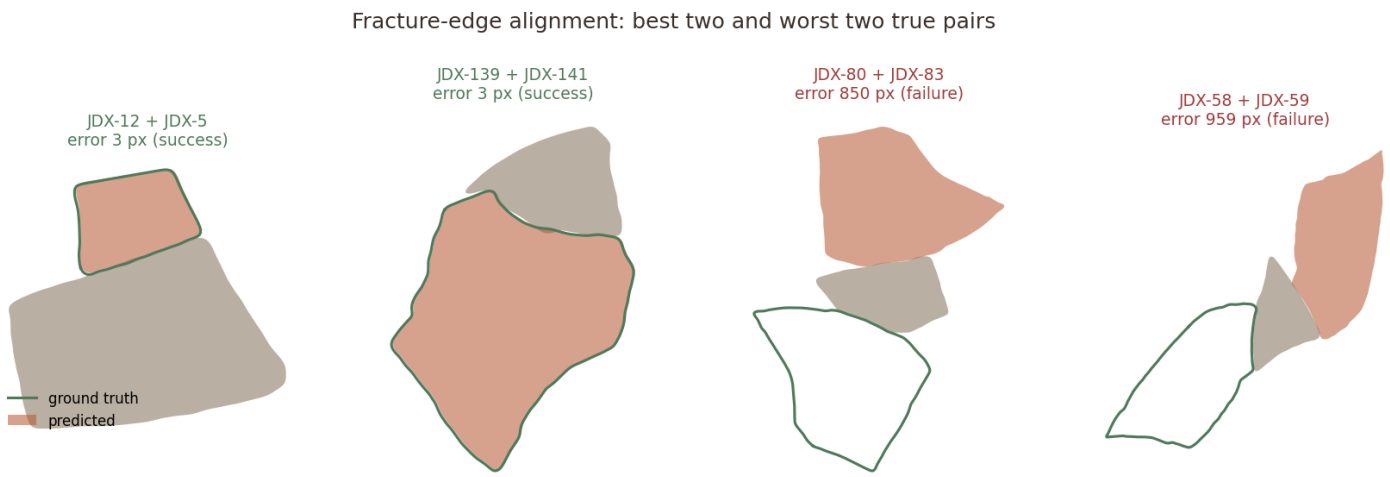
47%

clean synthetic breaks aligned within 50 px

17%

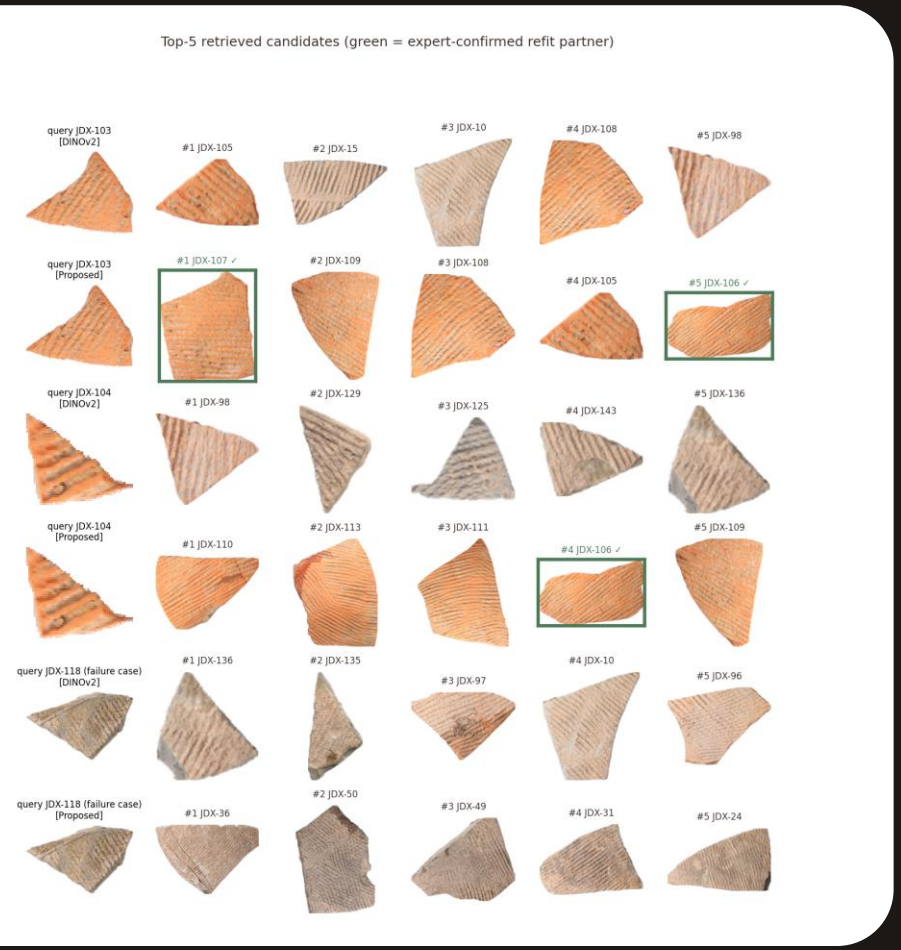
real eroded sherds aligned within 50 px

- At the expert pose, true partners touch along **87 px** (median)
- The best **wrong** pose found by ICP touches along **129 px**
- On worn edges, maximising contact chooses wrong fits



RESULTS ANALYSIS

Where it fails, and why



11%

of wrong top-1 picks are from the same vessel: the confusion is between vessels of one ware

7×

more same-vessel sherds in the top 5 than chance

0.44→0.71

Hit@5, smallest vs largest third of sherds

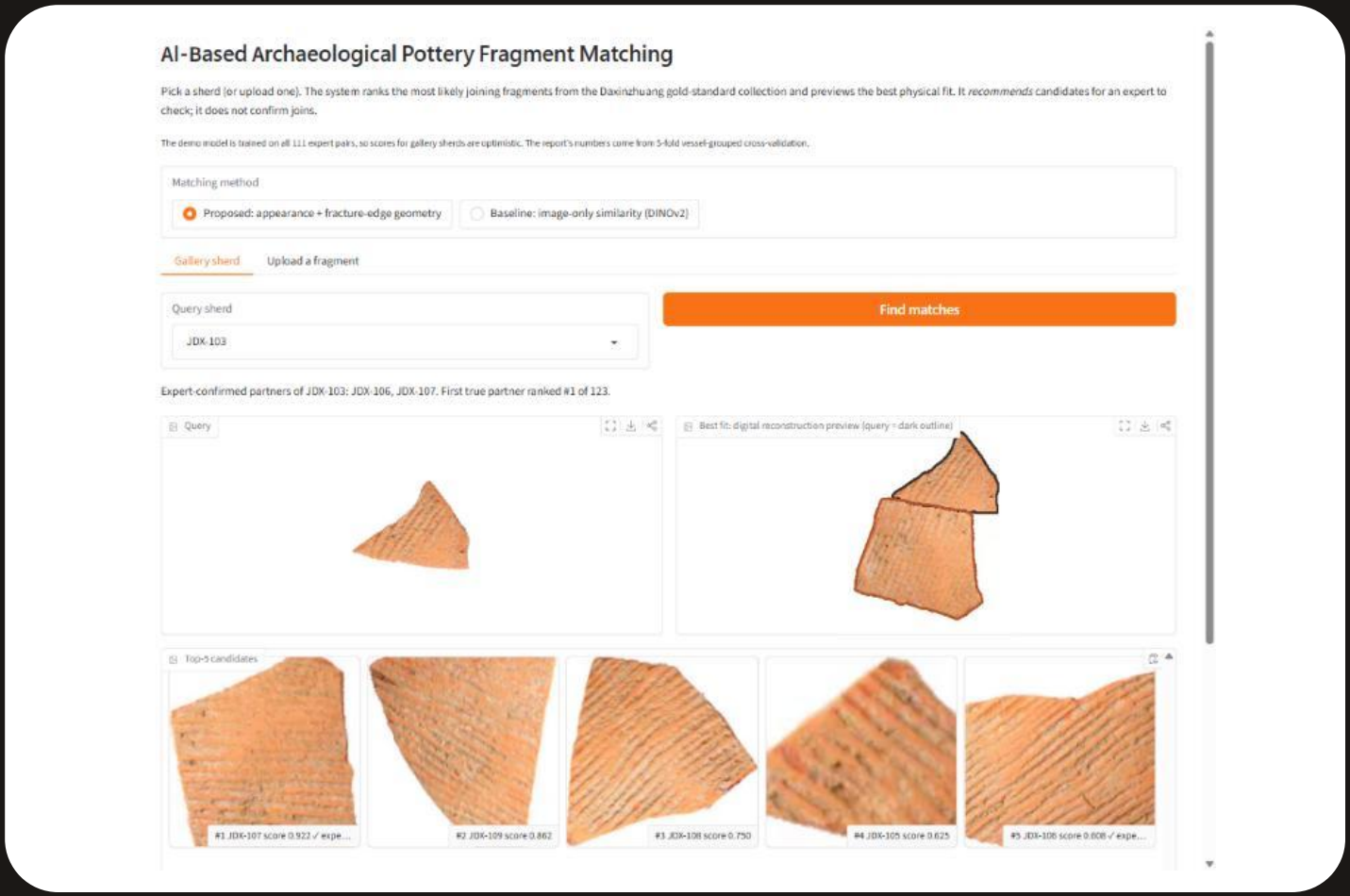
3.4×

PR-AUC over chance (precision 0.37, recall 0.23, F1 0.29)

Accuracy is 91.6%, but predicting 'no join' for everything already scores 92.8% on these test pairs, so it is not a useful measure here.

PROTOTYPE

Working demo



- Pick a sherd or **upload a photo**
- Top-5 candidates with scores
- **Evidence table**: why each was suggested
- **Fit preview**: first step to digital reconstruction
- Switch: proposed vs image-only

python app.py

github.com/vansh070605/pottery-fragment-matching

INFERENCE

What the evidence supports

USEFUL TRIAGE TOOL

An expert checking 10 suggestions finds a partner for 72% of sherds

H1 PARTLY SUPPORTED

Better than image-only, but the gain comes from fusing appearance cues

CLAIMS WE DO NOT MAKE

- Not an automatic reconstruction
- Not 'geometry is useless': 3-D and thickness untested
- One site, one ware

EDGES ARE THE BOTTLENECK

2-D outline geometry fails on eroded edges, not on clean ones

COLOUR MATTERS MOST

Vessels fired unevenly: colour identifies the vessel better than shape

LIMITATIONS & FUTURE SCOPE

What comes next

LIMITATIONS

- 2-D outlines miss thickness and wall curvature
- Max-contact pose choice fails on worn edges
- Only 111 positives: wide confidence intervals
- Exterior view only; one site and ware
- Raw-photo masks: shadows, light edges

Interior view + edge-band colour — LOW

independent evidence across the joint

Learned pose scoring — MEDIUM

train on expert vs false ICP poses

Contrastive fine-tuning — MEDIUM

better vessel grouping

Multi-piece assembly — HIGH

erosion-aware simulation → full reconstruction

REFERENCES

Sources and code

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Huang, Q.-X., et al. (2006). Reassembling fractured objects by geometric matching. ACM TOG, 25(3), 569–578.

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Kabsch, W. (1976). Acta Cryst. A32, 922–923 · Besl, P., & McKay, N. (1992). IEEE TPAMI 14(2), 239–256.

Full list (23 references): Review 2 case study report, Section 16.

Code, results, report

github.com/vansh070605/pottery-fragment-matching

Thank you

Questions?

